Some of the pros and cons of each approach are shown in table 1.1.

Table 1.1 The pros and cons of building your own solution versus buying off the shelf.

Option	Pros	Cons
Custom build	Ultimate flexibility; perfect alignment with unique business requirements; ensures complete data ownership	Requires extensive in-house technical expertise; takes longer to implement; demands ongoing maintenance
Buy off-the-shelf	Proven product; rapid deployment; reduced need for in-house development	Limited to existing features; requires adapting business processes to the software; difficult to differentiate your business; may compromise data sovereignty if operational data resides in vendor-controlled environments
Hybrid (build on PaaS / customization)	Faster time-to-market than full custom build; leverages existing framework for core functions; allows for critical customization to differentiate	Dependency on the platform vendor; customizations can be costly and create upgrade difficulties; requires a mix of in-house and vendor expertise

The decision is ultimately a strategic choice based on your short-term needs versus long-term goals, available technical expertise, how critical the digital twin will be to your organization, and your tolerance for external data dependencies. Given the breadth of technology involved, the most pragmatic approach is often a hybrid one of buying standard solutions for commodity components while building custom components that offer competitive differentiation or require strict data control.

1.6 Summary

- Recent advances in IoT, cloud computing, and AI/ML have made the technology required to build a digital twin widely available.
- A descriptive digital twin provides a static digital representation of reality.